

complexity of deploying new version of code on production environment is a difficult task which involves multiple processes and configurations.

Automation Advantages:

1. Repeatability
2. Time Saving
3. Quick Feedback
4. Change management

Practices of Automated Deployment:**Version is Bible**

Most deployment processes include a built-in safety net, preventing out-of-process, locally generated artifacts from entering the production environment. Every change a developer makes must be committed to a source control repository which is the only source for the build process.

This insistence on having a single source of truth—and a reliable one—creates a stable foundation for all development processes. However, the processes are only as strong as their weakest link. That's why the first golden rule for effective continuous delivery is to version-control everything. The process of versioning should not only be limited to code but include configuration, scripts, databases, website html and even documentation.

Build Binaries Once

The final build may be a single file or a complex build with several different possible deployment versions, each build version should be created in exactly the same way and result in unique artifacts. These artifacts should then be only copied and not rebuilt again when moving between environments.

This one-time-only compiling greatly reduces the risk of untracked differences due to various deployment environments, third-party libraries or different compilation contexts or configurations that will result in an unstable release going out.

We should save the compilation phase output (the binaries) to a binary repository, from which the deployment process can retrieve the relevant artifacts.

Smoke Test

Smoke testing is a very rapid way to make sure that the most crucial functions of a program are working as expected. This is non-exhaustive software testing that does not provide the same level of depth as full test suites but